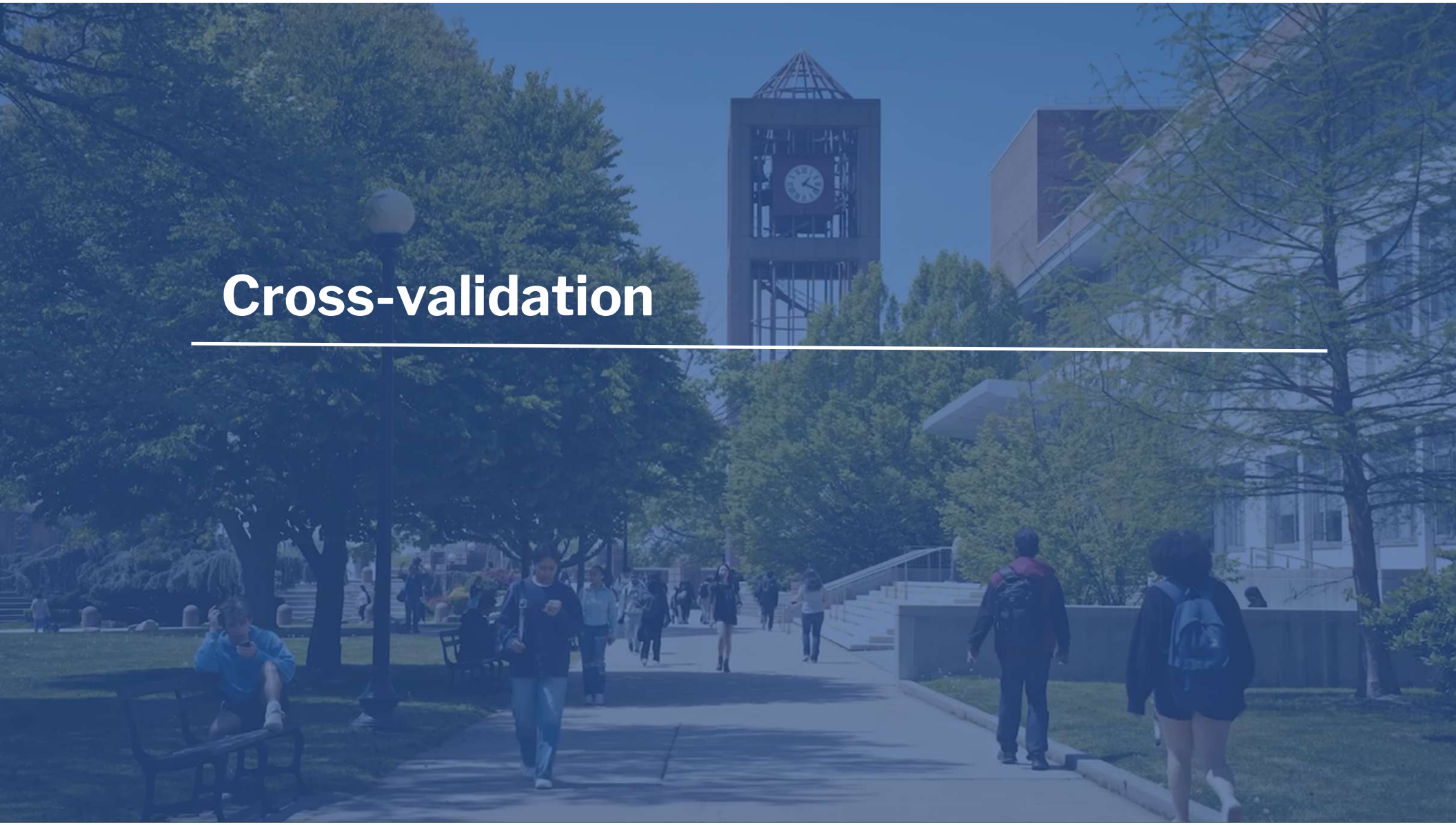
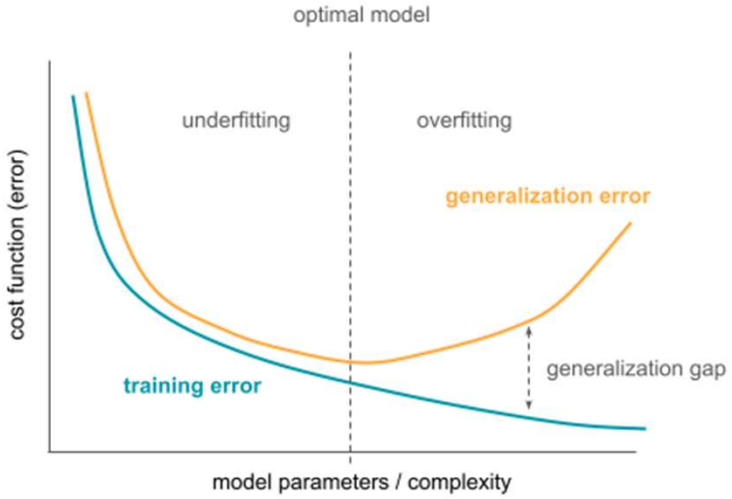

Introduction to Machine Learning
(GAI 601)
WEEK 10
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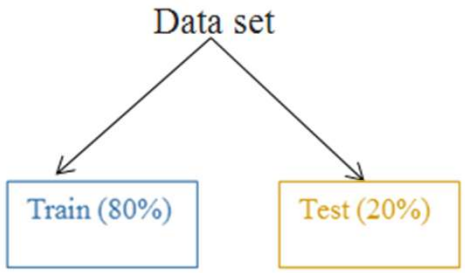

Cross-validation

Model Evaluation

Why model evaluation matters:

- **Training performance is misleading**
  A model can perform extremely well on training data but fail on new data due to overfitting. What matters in practice is performance on unseen data.
- **Goal: estimate real-world performance**
  The purpose of model evaluation is to approximate how the model will behave in deployment, not just on historical data.
- **Evaluation drives decision-making**
  In business settings, model selection impacts cost, risk, and outcomes, so evaluation must be rigorous and defensible.



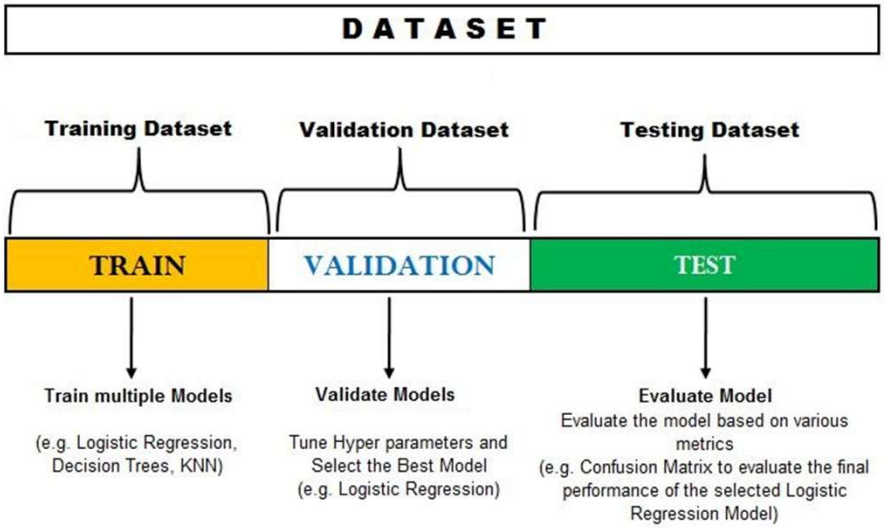


Source: SfB knowledge base

Train/Test Split

The benefits of cross-validation over a single test/training data split:

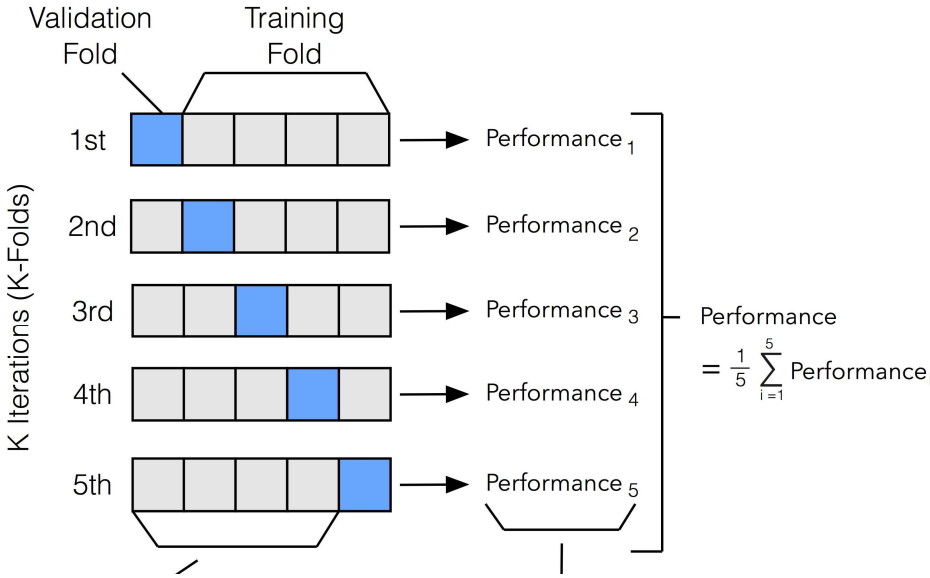
- **Reliability**
  Cross-validation evaluates the model across multiple train/test splits, reducing dependence on a single partition.
- **Robustness**
  By averaging results across folds, we obtain a more robust estimate of model performance.
- **Data optimization**
  Each observation is used for both training and validation, maximizing data efficiency.



Source: datavedas.com

k-fold cross-validation

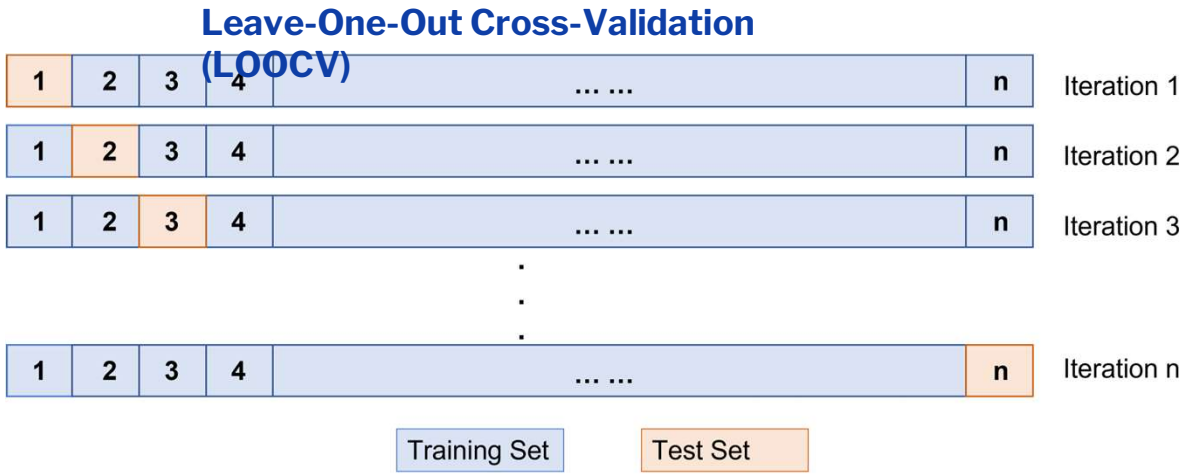
- **Training data is split into k-folds (subsets)**
  The dataset is divided into k equal parts, and each part is used once as a validation set.
- **Repeated runs improves performance**
  The model is trained k times, each time leaving out one fold for testing.
- **Values for k**
  Common values like 5 or 10 balance computational cost and stability of estimates.



Source: Sebastian Raschka, CCA4.0 License for educational purposes

Alternatives to k-fold

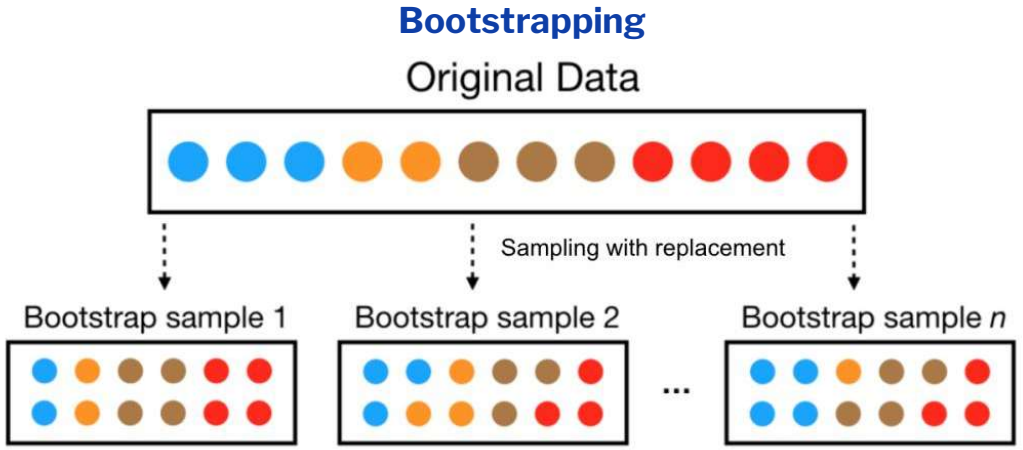
- **Leave-One-Out Cross-Validation (LOOCV)**
  Uses one observation for testing and the rest for training, leading to low bias but high variance.
- **Repeated k-fold**
  Running k-fold multiple times with different splits further reduces variance in estimates.
- **Stratified k-fold for classification**
  Ensures class proportions are preserved in each fold, which is critical for imbalanced datasets.



Source: rpubs.com

Bootstrap Resampling

- **Sampling with replacement**
  Bootstrap creates new datasets by randomly sampling observations with replacement, allowing duplicates.
- **Out-of-bag evaluation**
  Observations not selected in a sample can be used as a validation set, providing a built-in performance estimate.
- **Useful for small datasets**
  Bootstrap methods help estimate uncertainty when data is limited.



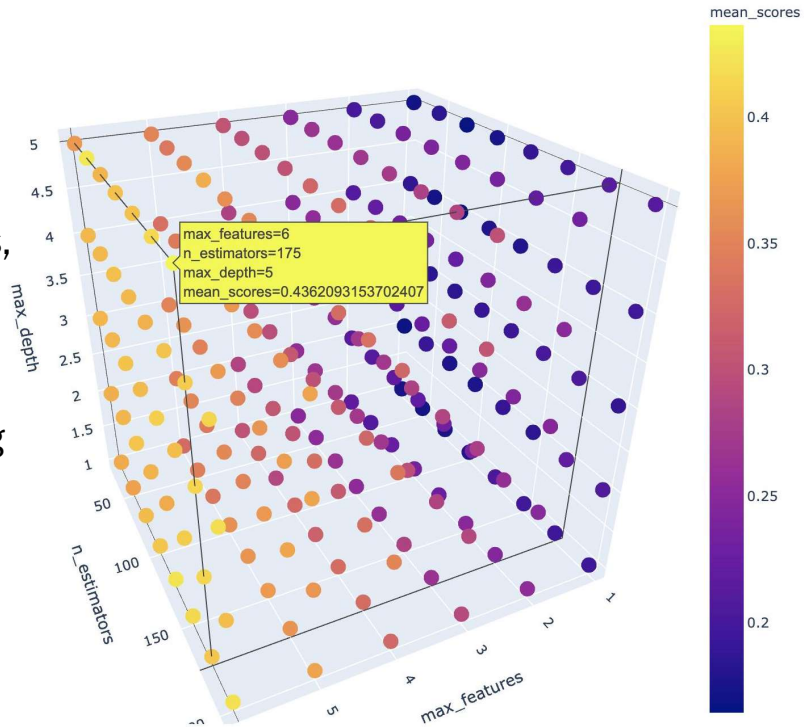
Source: aiml.com



Hyperparameter Tuning

Model Selection & Hyperparameters

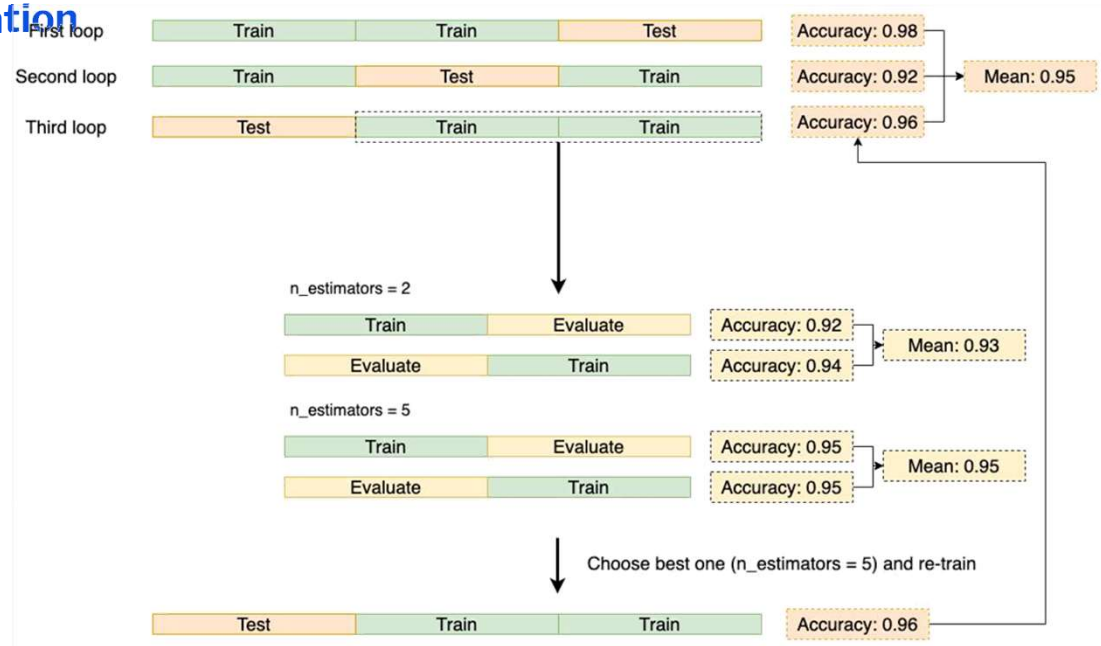
- **Models require tuning**
  Many models depend on hyperparameters that control complexity and behavior, such as tree depth or regularization strength.
- **Search strategies**
  Grid search systematically explores combinations, while random search is often more efficient in practice.
- **Risk of overfitting during tuning**
  Without proper validation, hyperparameter tuning can lead to overly optimistic results.



Source: dataasprirant.com

Nested Cross-Validation

- **Separates tuning and evaluation**
  Nested cross-validation uses an inner loop for hyperparameter tuning and an outer loop for performance estimation.
- **Prevents optimistic bias**
  This approach ensures that evaluation is not influenced by the tuning process.
- **Gold standard for rigorous evaluation**
  Particularly important in research and high-stakes applications.



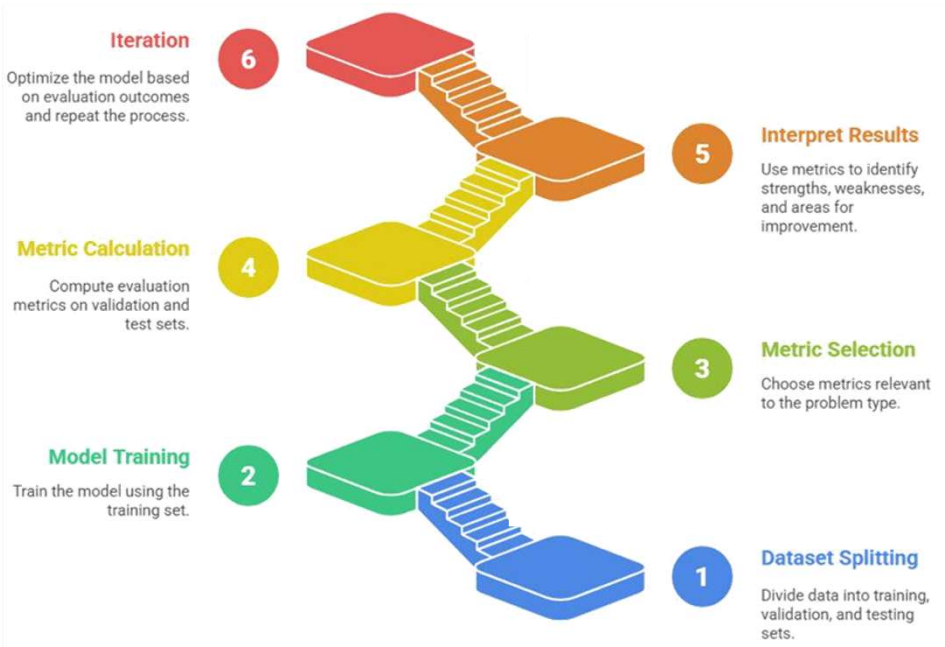
Source: dataasprirant.com



Metrics

Choosing the Right Metric

- **Metrics reflect business goals**
  The choice of metric should align with the cost of errors in the real-world application.
- **No universal best metric**
  Different tasks and domains require different evaluation criteria.
- **Threshold-dependent vs independent metrics**
  Some metrics depend on classification thresholds, while others (like AUC) evaluate ranking performance.



Source: dataasprirant.com

Confusion Matrix, ROC & AUC

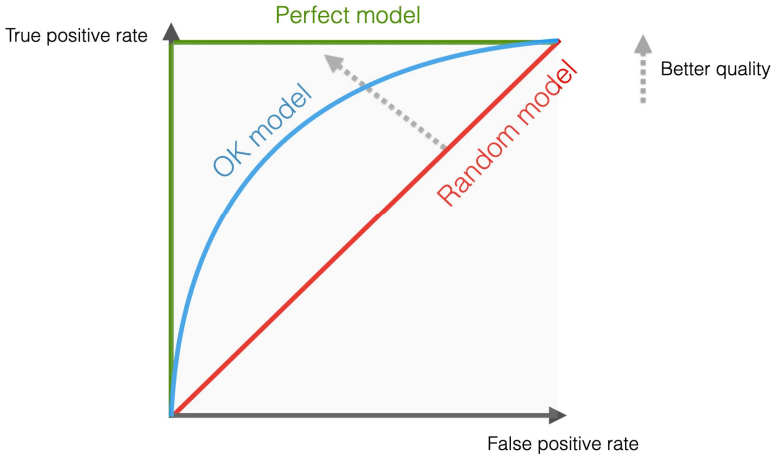
Confusion Matrix

- A table that used to evaluate the performance of a classification model.
- It provides a comprehensive summary of the model's predictions compared to the actual outcomes in a tabular format.
- Each row of the matrix represents the instances in a predicted class, While each column represents the instances in an actual class.

	Predicted Positive	Predicted Negative	
Actual Positive	TP *True Positive*	FN *False Negative*	Sensitivity $\frac{TP}{(TP+FN)}$
Actual Negative	FP *False Positive*	TN *True Negative*	Specificity $\frac{TN}{(TN+FP)}$
	Precision $\frac{TP}{(TP+FP)}$	Negative Predictive Value $\frac{TN}{(TN+FN)}$	Accuracy $\frac{TP+TN}{(TP+TN+FP+FN)}$

Receiver Operating Characteristic (ROC) Curve

- A graphical representation of the trade-off between the TP rate and the FP rate for different threshold values.

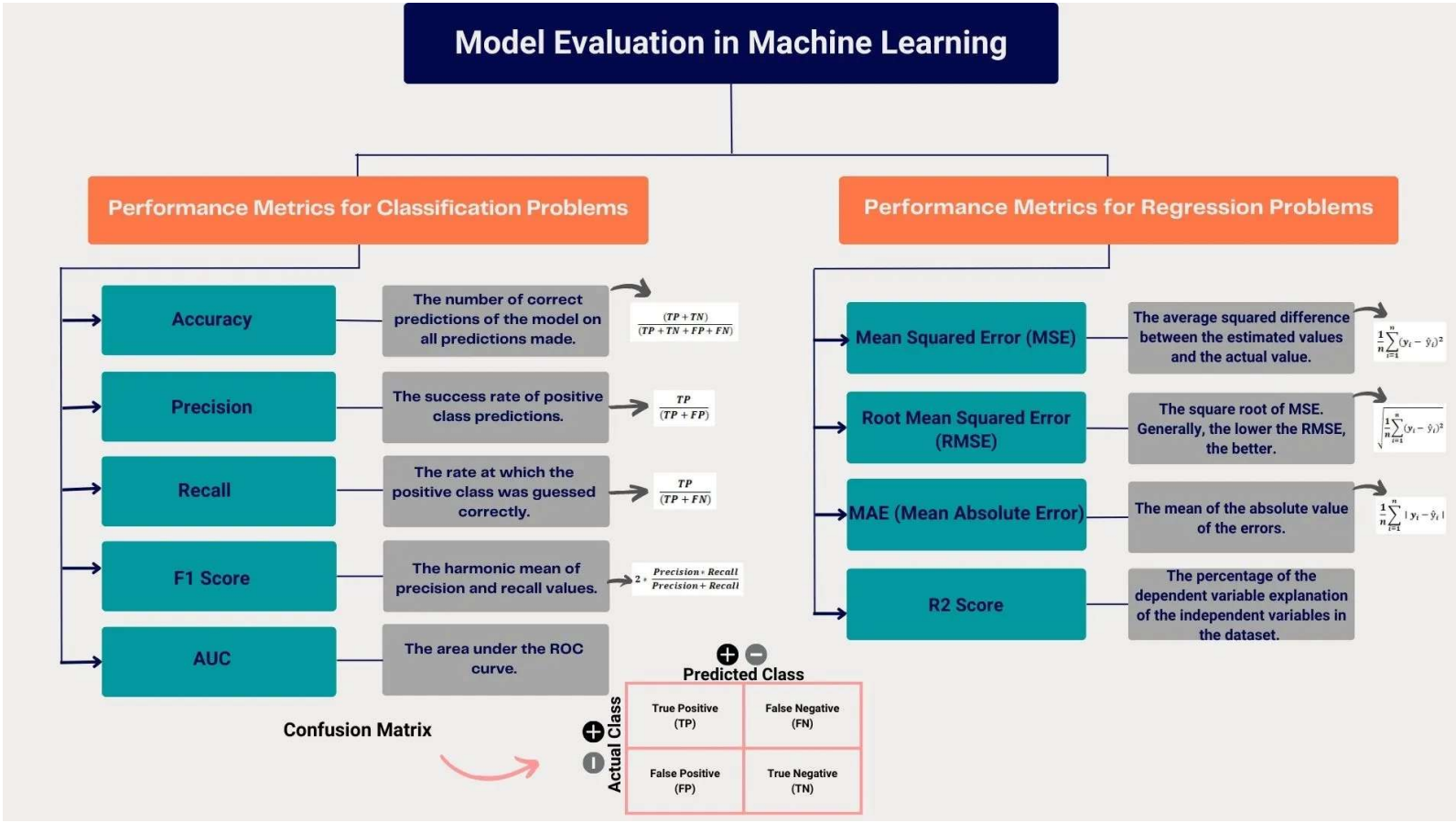


Area under the ROC curve (AUC)

- A number that describes the overall performance of a classification model by calculating the area under the ROC curve.

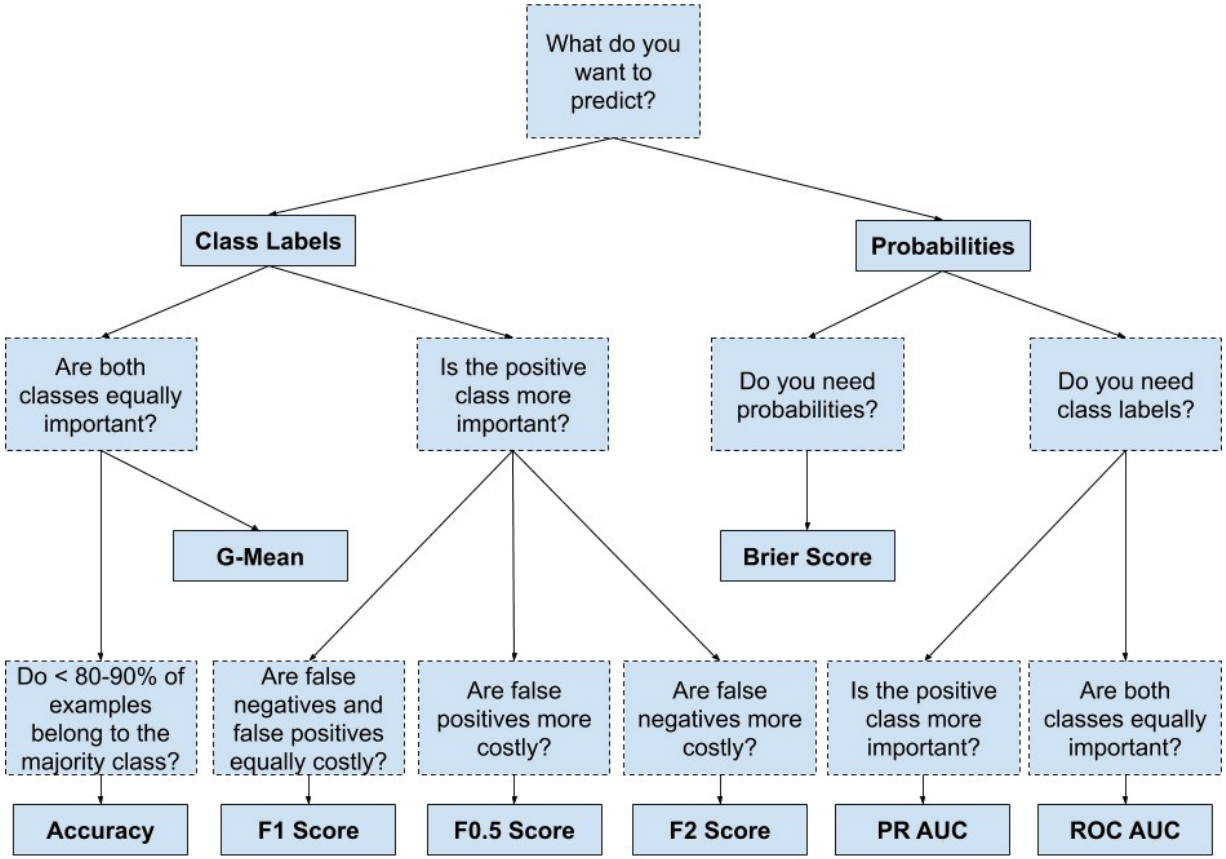
"Source: Integrating image filters and confidence thresholds to improve cropland data layer for deep learning–based cropland segmentation", Reza Maleki et al, Feb 2026

Classification & Regression metrics

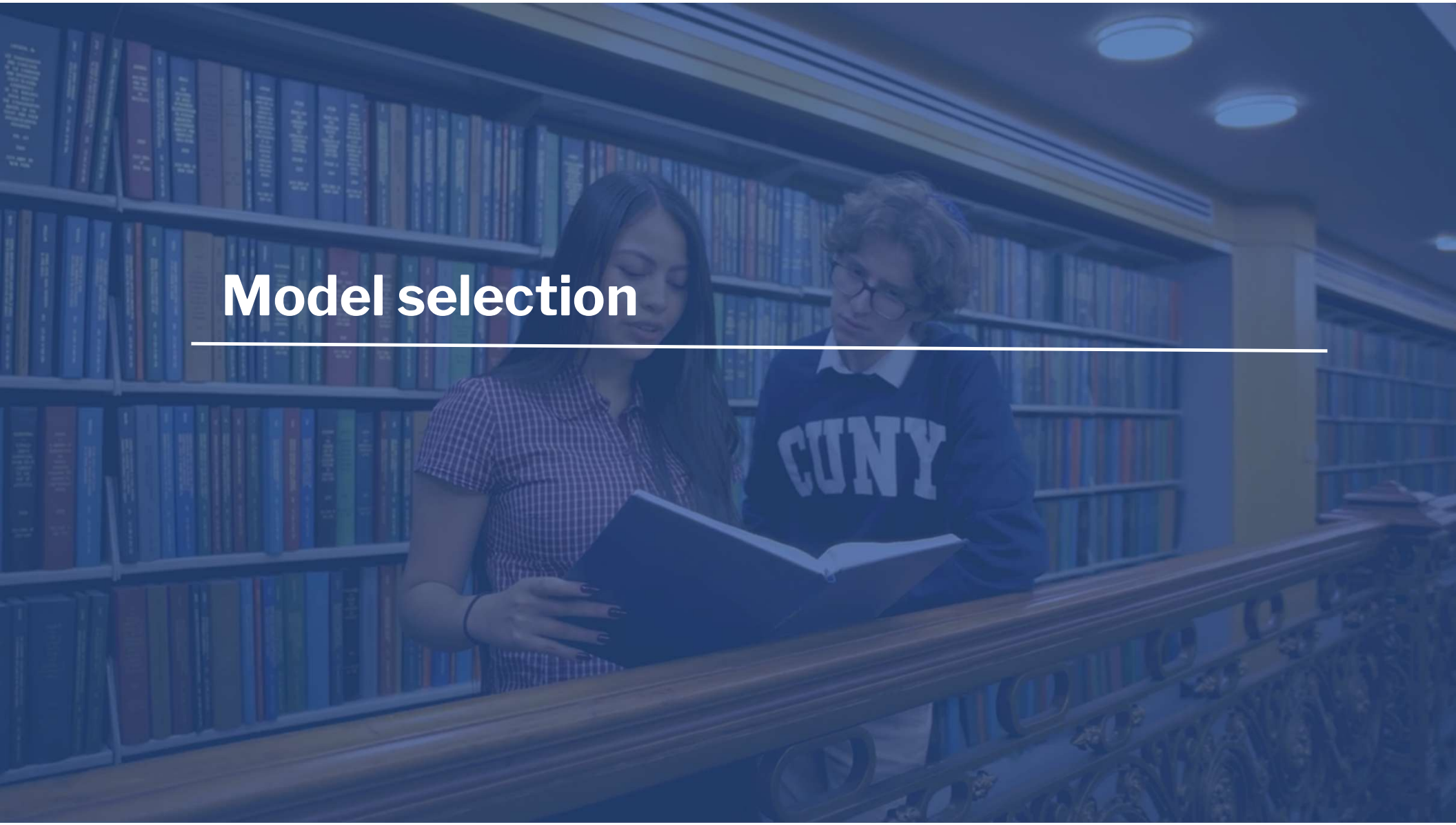


Source: Medium - N. Avsar

Choosing the right metric



Source: Machine Learning Mastery



Model selection

No free Lunch theorem

No-Free-Lunch Theorem states

- No single classifier works the best for all possible problems
- We need to make assumptions to generalize (we need bias)



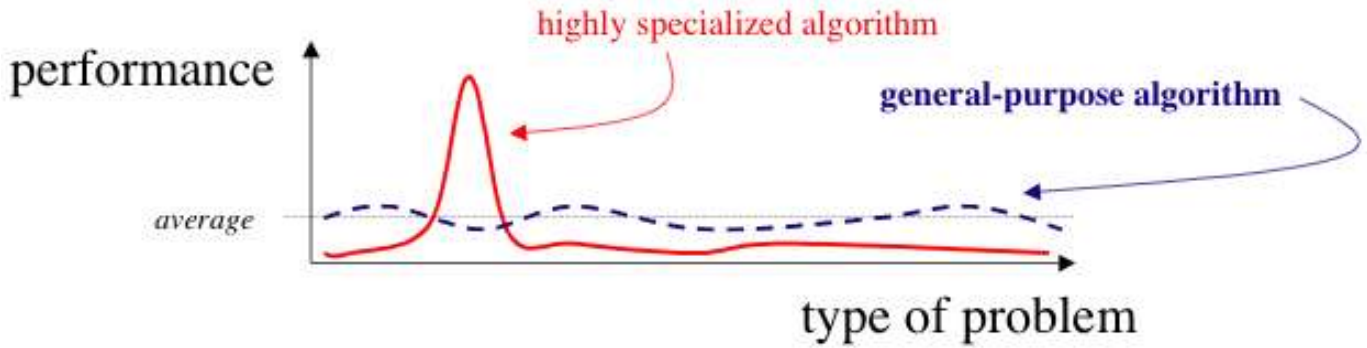
Source: Wikipedia

No free Lunch theorem

Theorem:

The average performance of any pair of algorithms across all possible problems is identical.

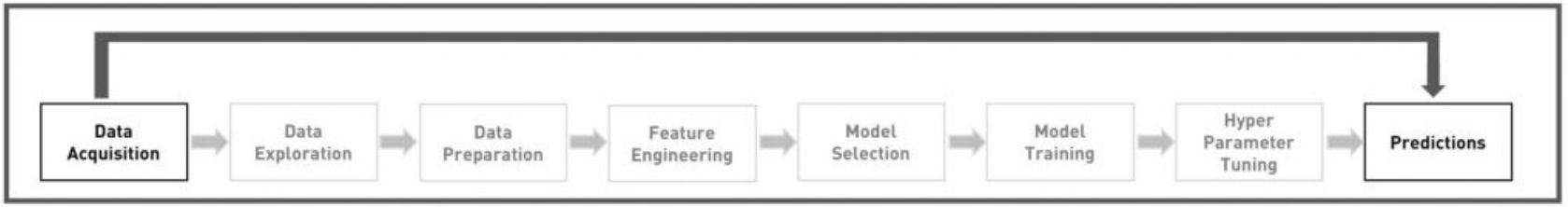
If an algorithm achieves superior results on some problems, then it must pay with inferiority on the other problems



Source: https://medium.com/@LeonFedden/the-no-free-lunch-theorem-62ae2c3ed10c

Automated Machine Learning (AutoML)

Can Machine Learning perform Machine Learning to identify optimal models?

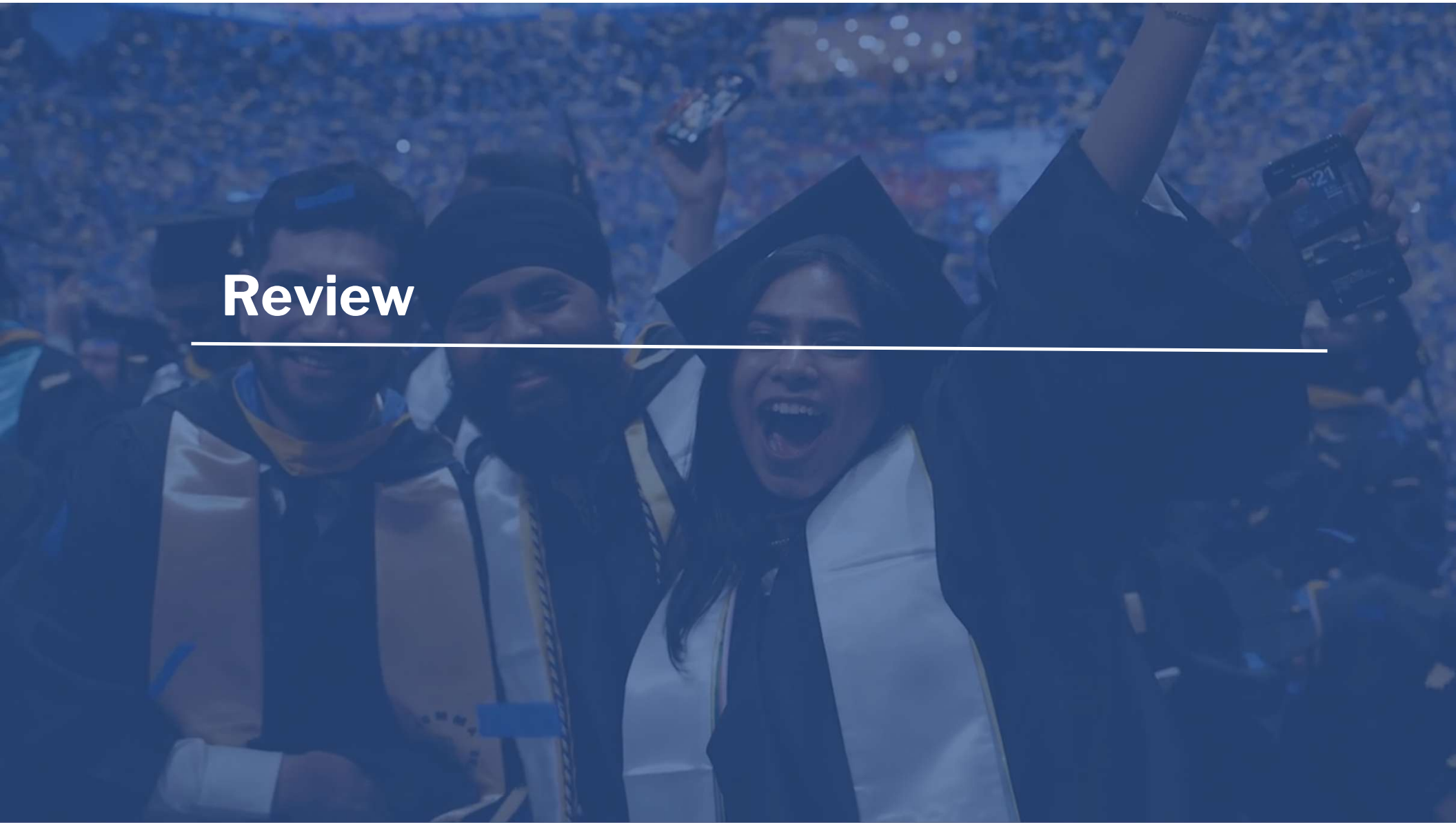


Let's consider the no-free lunch theorem again.

Q: If we automate the ML Lifecycle with AutoML, does that mean no-free lunch doesn't apply?

A: No, no-free lunch theorem still applies because:

- AutoML is just optimizing the process – it is not doing creative work
- AutoML will give you faster results – but not necessarily better (optimal) results



Review

This week we covered

Lesson Objectives/Topics

- **Foundational**
  - Apply model selection methods such as k-fold cross-validation.
  - Evaluate resampling techniques to estimate performance on unseen data.
  - Select appropriate metrics (AUC, F1-score) for different ML tasks.
  - Evaluate mode
- **Strategic**
  - AutoML & the no free lunch theorem


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